

DEPARTMENT OF INFORMATION TECHNOLOGY

IT3708 - BIO-INSPIRED ARTIFICIAL INTELLIGENCE

Project 3 - Understanding Optimization Landscapes using Bio-inspired Algorithms

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1 Introduction

This project studies how optimization landscape structure affects search performance. I compare three algorithm families, a single-objective evolutionary algorithm (SGA), a multi-objective evolutionary algorithm (NSGA-II), and a swarm-intelligence method (ACO), on HDF5-based feature-selection landscapes and a synthetic triangle benchmark. For each landscape, I analyze penalized fitness, identify local optima, and use repeated runs to compare convergence and robustness under the same settings. The goal is to show why different problem structures favor different search strategies and why no single optimizer is best for every landscape.

A summary of how project goals map to code is given in Table 2.

2 Algorithms

The three methods were implemented in a common pipeline so outputs are directly comparable.

2.1 Single-objective algorithm (SGA)

SGA is a generational binary GA with tournament selection, one-point crossover, bit-flip mutation, and elitism; it records best-so-far convergence per run.

2.2 Multi-objective algorithm (NSGA-II)

NSGA-II optimizes accuracy, feature count, and time using Pareto dominance, non-dominated sorting, crowding distance, and parent-offspring elitist selection.

2.2.a Ant colony optimization algorithm (ACO)

ACO samples binary solutions from pheromone-heuristic probabilities, mixes guided and random exploration, applies local search to top ants, and updates pheromone with evaporation, bounds, and stagnation reset.

3 Method

I compare SGA, NSGA-II, and ACO on two landscape types: precomputed feature-selection landscapes from HDF5 files and a synthetic triangle benchmark ($n = 16$, $m = 1$, $s = 4$). For feature selection, fitness is computed from averaged accuracy with penalties for selected-feature count and normalized training time.

To keep the comparison fair, all methods use the same experiment budget: 10 seeds (1000-1009), population/ant size 120, and 150 generations or iterations. Shared GA operators are tournament selection (size 3), one-point crossover (0.85), and bit-flip mutation (0.02).

3.1 Synthetic landscape triangle function

The triangle function is implemented following [1] and [2] p. 23. Fitness depends only on the number of active bits, producing a regular landscape that is useful for controlled optimizer comparison.

3.2 Hamming-1 distance calculation

In an n -bit space, each solution has exactly n Hamming-1 neighbors. This matches single-bit mutation and is the neighborhood used to detect local optima. A bitstring is locally optimal if its fitness is higher than all one-bit-flip neighbors.

3.3 Run configuration

Runs are configured through `configuration.json`, loaded by `main.jl`, and executed with optimizer-specific parameters (`sga`, `nsga-ii`, `aco`). Each run writes best fitness, convergence traces, and optional landscape/optima plots.

3.4 Visualization

The visualization scripts generate landscape plots, 3D animations, and convergence charts so the structure of each problem and the behavior of each optimizer can be inspected visually.

4 Results

This section tries to summarize some of the experiment outputs.

4.1 Feature-selection landscapes

For the three feature-selection landscapes (`breast_w_01`, `letter_r_05`, and `credit_a_08`), ACO and NSGA-II consistently outperformed SGA in terms of best fitness and mean best fitness over 10 seeds, see Table 1.

Overall, ACO and NSGA-II reached the same best values on all three feature landscapes, while SGA converged to clearly lower-quality solutions in this configuration.

4.2 Synthetic triangle landscapes

For the main synthetic setting ($n = 16$, $m = 1$, $s = 4$), both ACO and NSGA-II reached the maximum observed fitness across all runs:

- Local optima (strict Hamming-1): 3640
- ACO: best/mean 4.0, std 0.0
- NSGA-II: best/mean 4.0, std 0.0

An additional smaller synthetic run ($n = 8$, $m = 1$, $s = 2$) with ACO also reached its top observed value consistently:

- Local optima (strict Hamming-1): 56
- ACO: best/mean 2.0, std 0.0

Dataset	Optimizer	Runs	Best	Mean	Std	Best bitstring	Loc opt
01	SGA	10	0.91473430	0.91473430	0.0	010001000	7
01	NSGA-II	10	0.97012195	0.97007622	0.00014	111001111	7
01	ACO	10	0.97012195	0.97012195	0.0	111001111	7
05	SGA	10	0.84103182	0.84103182	0.0	000000001000000	1
05	NSGA-II	10	0.89126275	0.88826530	0.0016	001110001110111	1
05	ACO	10	0.89126275	0.89123086	0.0001	001110001110111	1
08	SGA	10	0.77858575	0.77858575	0.0	0000000111111011	2
08	NSGA-II	10	0.95633334	0.95633334	0.0	0000011111111111	2
08	ACO	10	0.95633334	0.95633334	0.0	0000011111111111	2
06	SGA	10	0.86631443	0.86433337	0.0042	0001010010001000	136
06	NSGA-II	10	0.97916666	0.97729166	0.0016	0101010011001001	136
06	ACO	10	0.97916666	0.97822916	0.0015	0101010011001001	136
10	SGA	10	0.84823218	0.84728136	0.0012	000001000000000011	773
10	NSGA-II	10	0.93749999	0.92890625	0.0063	010110100000000010	773
10	ACO	10	0.93229167	0.92981771	0.003	010101100000000010	773

Table 1: Batch experiment summary across all feature-selection landscapes and optimizers.

These results indicate that both ACO and NSGA-II handle the structured triangle landscape effectively, even when the number of local optima is large.

4.3 New test data with large synthetic landscape

The new asymmetric triangle function has a single global optimum at 31 active bits. Since the NSGA-II algorithm chosen to optimize this landscape rewards fewer bits, search pressure is towards the opposite end of the landscape.

4.4 Comparative interpretation

Two patterns are clear from the artifact summaries:

- Algorithm ranking: ACO and NSGA-II were the strongest methods on all reported feature-selection landscapes, with practically identical best performance.
- Robustness: Most runs had very low dispersion across seeds, and several configurations achieved zero standard deviation, suggesting highly stable convergence under the selected parameter settings.

At the same time, the 05-credit-a landscape showed slightly larger variance for NSGA-II than for ACO, which may indicate higher sensitivity to initialization or parent-offspring selection dynamics on that specific landscape.

4.5 Batch experiment summary

The batch experiment summary below aggregates the feature-selection runs across the three landscapes and all three optimizers.

4.6 Comments on the included figures

- Figure 1 shows the breast-w-01 landscape after ACO has identified 9 local optima. With a landscape that is relatively small, peaks are easily identifiable. Even so, some of the optima are hard to read due to overlapping text.
- Figure 2 shows how the NSGA-II algorithm converges on the same dataset, with the best result discovered around generation 140.
- Dataset letter-r-05 has a lot more datapoints, as shown in Figure 3. It's not really feasible to make out details, but the final optima is still plotted.

Lab goal	Where it is covered	Primary files
Read and parse large evolutionary landscapes based on feature selection, and implement a synthetic landscape.	HDF5 parsing reads accuracies and times from file, builds feature-selection lookup fitness. Synthetic triangle landscape and triangle fitness is implemented.	src/feature_landscape.jl src/ triangle_landscape.jl
Create a visualization of the fitness landscape to highlight optima and show complexity.	Local optima are detected with Hamming-1 neighborhood check. Landscape plots highlight local optima and summarize fitness structure.	scripts/ visualize_landscape.jl scripts/ animate_landscape_3d.jl
(Optional) Visualize algorithm behavior on search spaces.	Convergence behavior is visualized from mean best-so-far traces across runs.	scripts/ visualize_convergence.jl
Implement one single-objective evolutionary algorithm.	SGA with tournament selection, one-point crossover, mutation, and elitism.	src/sga.jl
Implement one multi-objective evolutionary algorithm.	NSGA-II with Pareto dominance, non-dominated sorting, crowding distance, elitism.	src/nsga2.jl
Implement one swarm intelligence optimization algorithm.	Ant Colony Optimization with pheromone/heuristic sampling, elite updates, local search refinement, and stagnation reset.	src/ant_colony.jl
Compare algorithms and show experimentation results.	Algorithm results are exported to CSV and summarized; result artifacts are generated per dataset/optimizer and discussed in this report.	src/main.jl artifacts/*/summary.md artifacts/*/runs.csv
Test algorithms on unseen landscape and try to find multiple optima.	Built-in parameterization of batch experiments. Configuration for unseen landscapes in run_zoo_hep_batch.sh. The same pipeline can be run on unseen HDF5 landscapes by changing dataset_path.	configuration.json

Table 2: Lab goals and where they are covered in code.

- I attempted a scatterplot of the same dataset in Figure 4, with fitness as a function of the number of active features and normalized time. I also produced an animation of this plot, rotating around the fitness axis, revealing interesting artifacts of the data that cannot be seen at this angle.
- In Figure 5 I visualize how the number of active features are distributed in the model. The local

optima shown in the lower histogram are all on 4 active features.

- On the triangle landscape shown in Figure 6, the local optima are more visible.

5 Appendix: Lab goals

This appendix maps each required lab goal to the implementation in the codebase, so the coverage can be verified directly from source files, see Table 2.

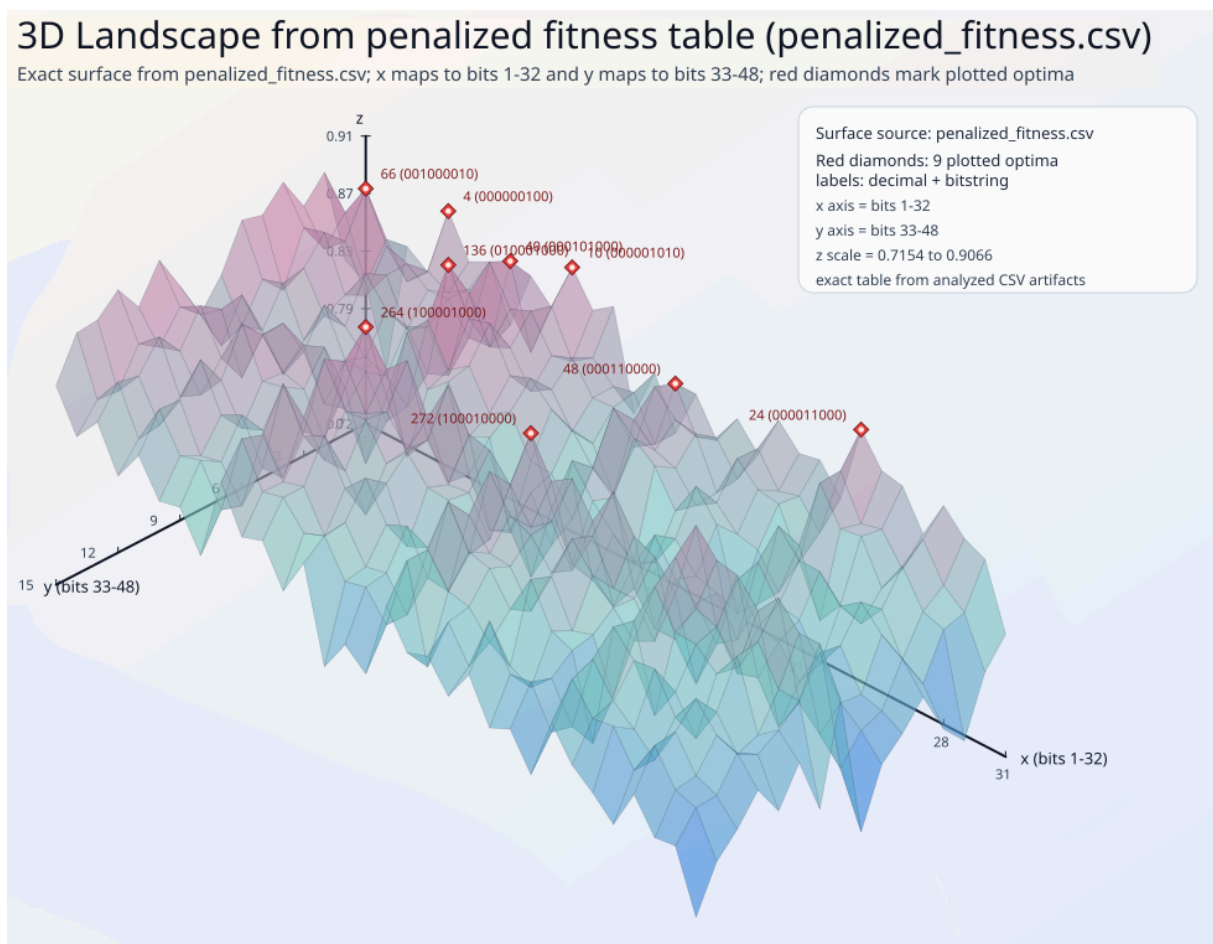


Figure 1: 3D fitness landscape from ACO on dataset 01.

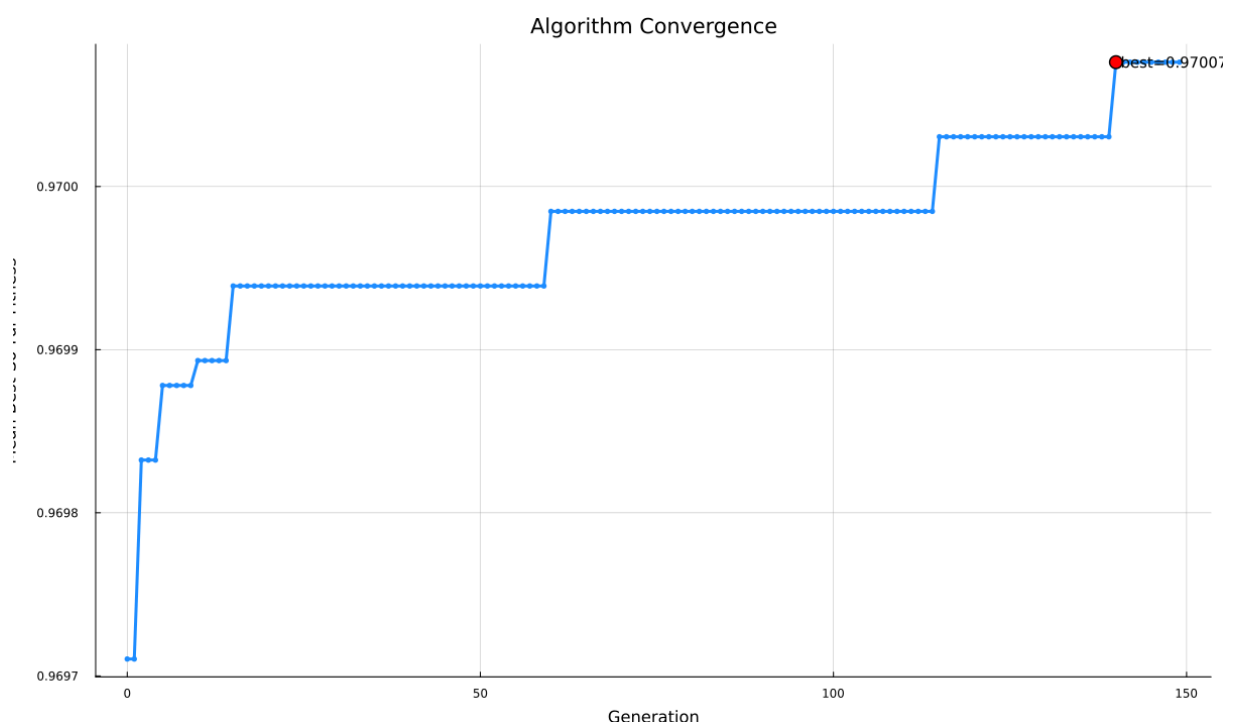


Figure 2: NSGA-II convergence curve for dataset 01.

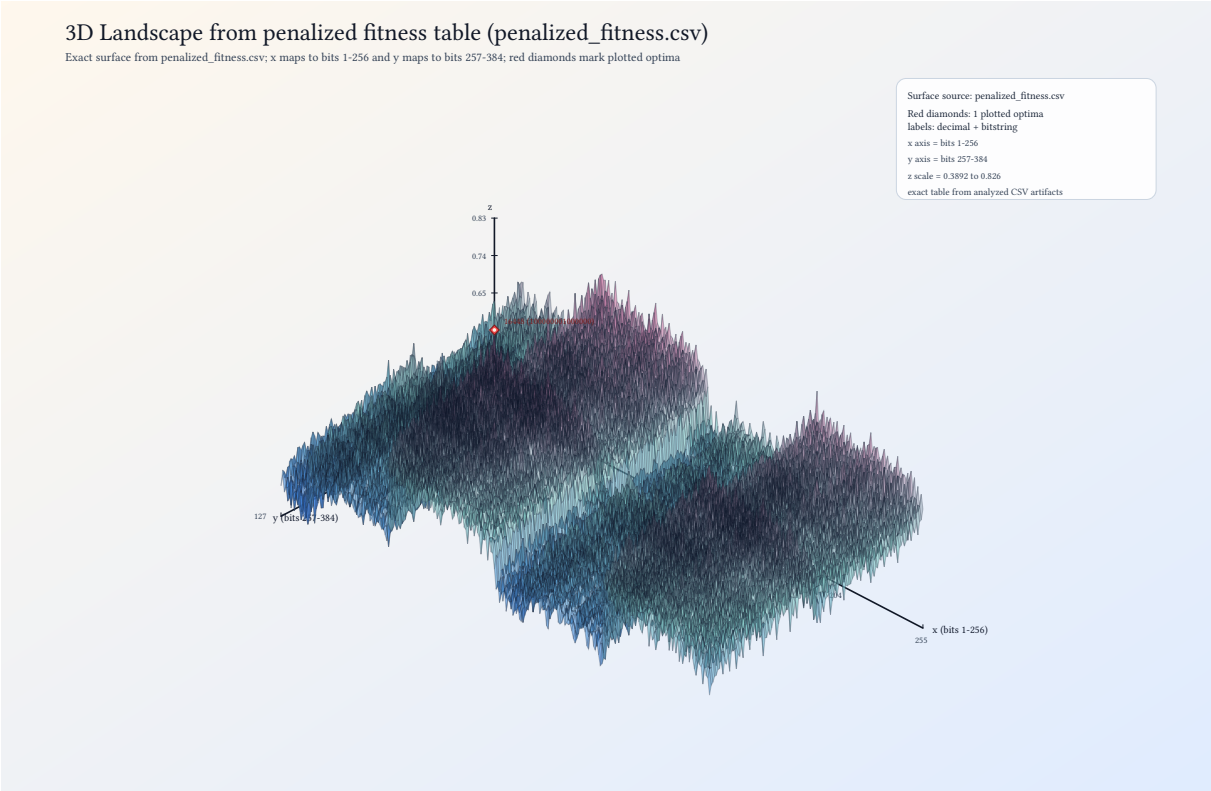


Figure 3: 3D fitness landscape from ACO on dataset 05.

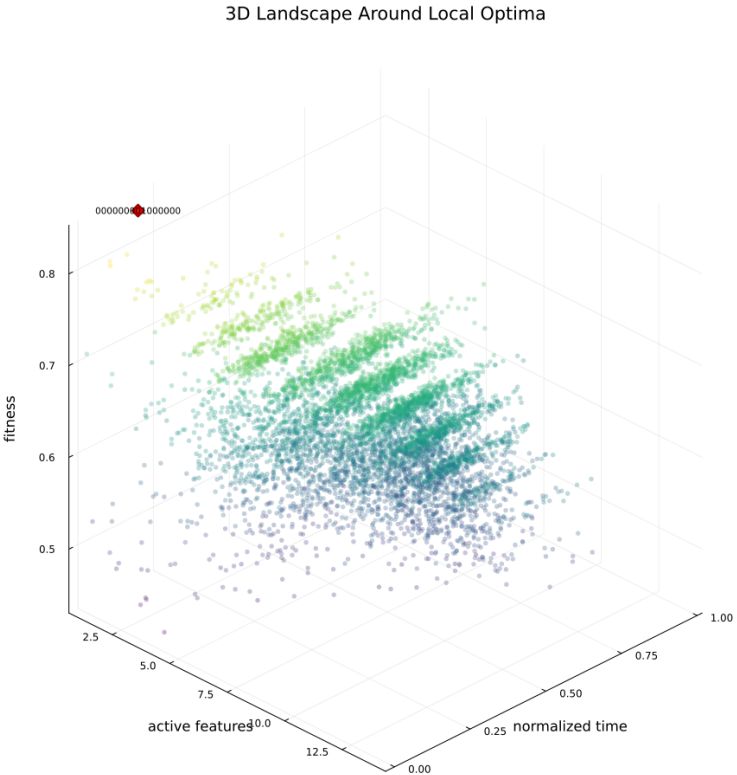


Figure 4: 3D scatterplot of the landscape using NSGA-II on dataset 05.

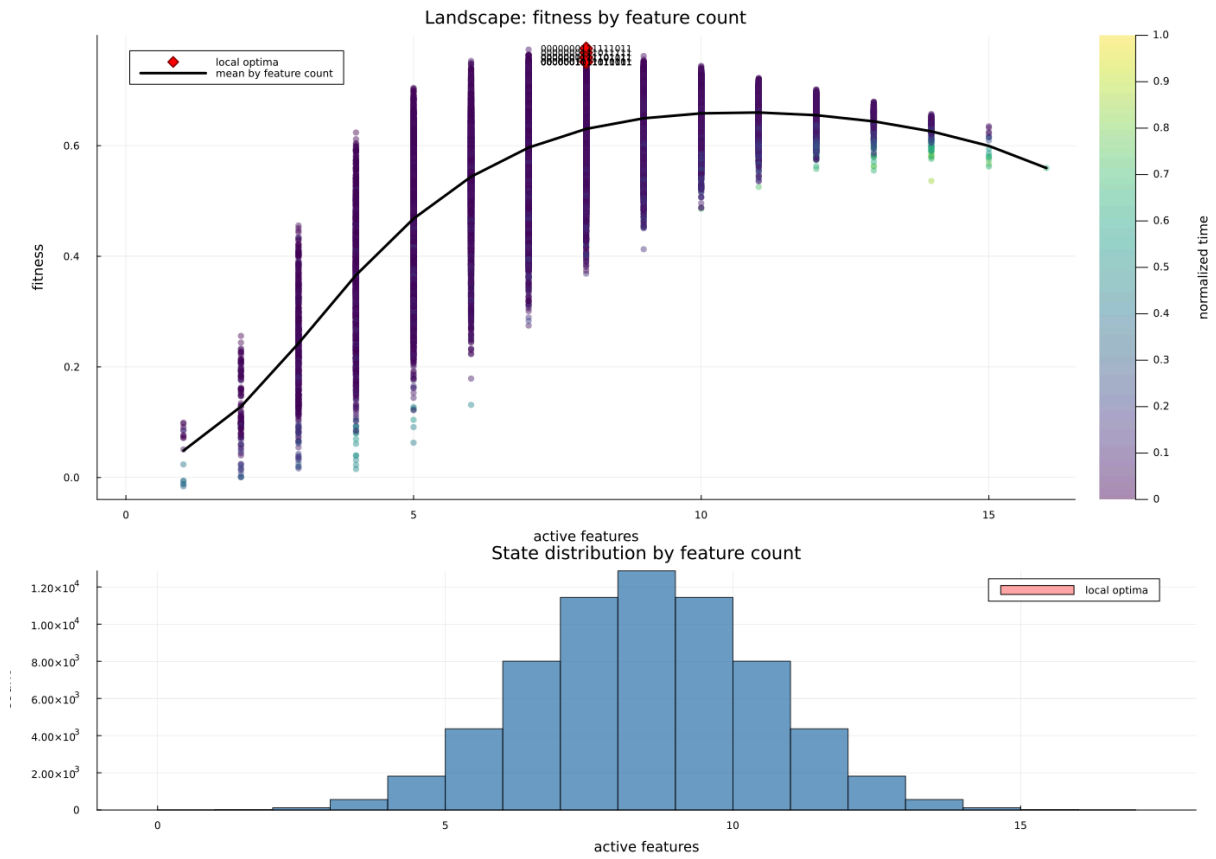


Figure 5: NSGA-II fitness distribution plot for dataset 08.

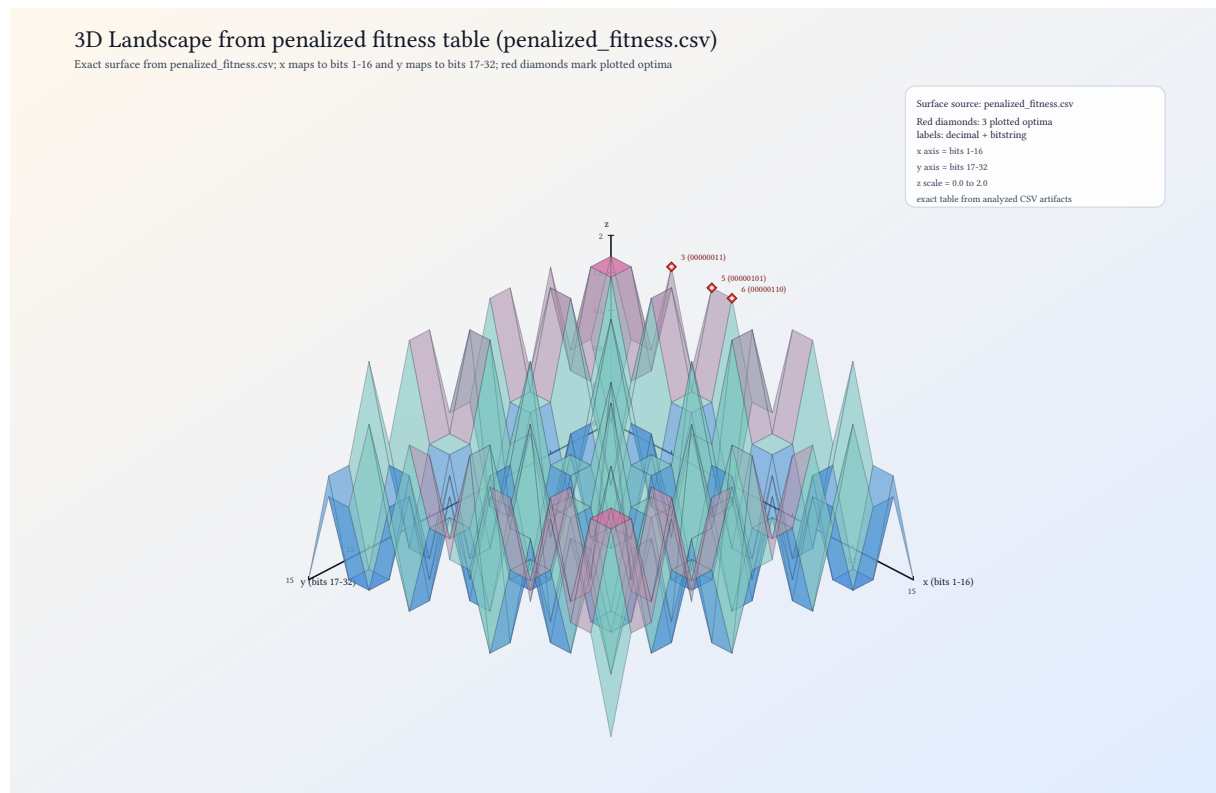


Figure 6: 3D fitness landscape from ACO on the triangle landscape.

6 Appendix: Images

Collection of images produced by the visualisation logic.

Bibliography

- [1] X. F. C. Sánchez-Díaz and O. J. Mengshoel, “Estimating the Number of Local Optima in Multimodal Pseudo-Boolean Functions: Validation via Landscapes of Triangles,” in *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, Melbourne, VIC, Australia: Association for Computing Machinery, July 2024, pp. 1–4. doi: 10.1145/3638530.3654156.
- [2] J. Å. Bergquist and X. F. C. Sánchez-Díaz, “Bio-AI Assignment 3: Understanding Optimization Landscapes using Bio-inspired Algorithms.” 2025.